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Special Collection: Atomic Layer Deposition (ALD)

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## ABSTRACT

Molybdenum nitride thin films are deposited using a newly synthesized liquid Mo precursor [$MoCl_4$(THD)(THF)] in an ALD super-cycle process. The new precursor is synthesized using $MoCl_5$ and 2,2,6,6-tetramethyl-3,5-heptanedione, which is a bidentate ligand. The synthesized precursor exists in the liquid phase at room temperature and has a characteristic of evaporating 99% at 150–220 °C. Using this new precursor in an ALD super-cycle process results in a pure $MoN_x$ thin film with few impurities (C and O). In addition, such $MoN_x$ thin films have relatively low resistivity values due to excellent crystallinity and a low impurity concentration. The films' diffusion barrier characteristics confirm that they can perform the role of a barrier at over 600 °C.





## I. INTRODUCTION

Currently, as semiconductor devices, such as memory and logic devices, become ultra-highly integrated, they are also being scaled down.[1,2] As the scaling down of semiconductor devices progresses, various problems occur; a representative problem is an increase in the resistivity of thin metal films. According to previously published research, as the thickness of the metal thin film decreases, the resistivity of the thin film increases exponentially for various reasons, such as electron scattering at external surfaces and grain boundaries.[3,4] In response to this problem, studies aiming to limit the increase in resistivity resulting from reductions in thin film thickness are in progress.

Among many candidate metal materials, a molybdenum metal (carbide) thin film has been in the spotlight as a next-generation metal thin film.[5] Although molybdenum is thin, its relative resistivity is low and its price is cheaper than precious metals such as gold, silver, ruthenium, and rhodium; so it is spotlighted as a metal that can replace tungsten and copper. However, Mo thin films have the disadvantages of long incubation times (a characteristic of a metal thin film) and high surface roughness due to island-type deposition.[6–8] In order to solve these problems, scientists seek to fabricate $MoN_x$ thin films that can be used as a seed layer. When a seed layer is used, the incubation time is insignificant when depositing the metal thin film, and the resulting roughness of the final metal film is low.[9,10] $MoN_x$ thin films being studied as seed layers have low resistivity values and can be deposited using the same precursor as the Mo thin film, thus increasing the efficiency of the process.[11–13] $MoN_x$ thin films have been deposited using PVD (physical vapor deposition) and CVD (chemical vapor deposition) processes. $MoN_x$ thin films deposited using the PVD process have few impurities, and the film quality is excellent, but unfortunately, the step coverage is subpar. Depositing a $MoN_x$ thin film using the CVD process is problematic in that a relatively high temperature process is required and the impurity content in the thin film is high. In order to solve these problems, an ALD (atomic layer deposition) process is in the spotlight. An ALD process enables large-area deposition on a substrate and allows for accurate thickness control. Additionally, the process temperature and the impurity content in the deposited thin film are low. Additionally, as the thin film is deposited in successive atomic layers, the ALD process provides excellent step coverage.[14–17] Therefore, many researchers are using the ALD process to deposit $MoN_x$ thin films. Additionally, even if the ALD process is performed at a temperature similar to that of the CVD process, fine thickness control is possible and the impurity content is low.

Currently, $MoCl_5$ and $Mo(CO)_6$ complexes have been widely used as precursors to deposit Mo films by the ALD process.[18–20] However, several problems exist when current precursors (e.g., $MoCl_5$ and $Mo(CO)_6$) are used as solid-state forms; (1) spontaneous intermolecular reactions would occur in the process of vaporization and transport to the gas phase, resulting in multicomponent compounds. (2) It is difficult to control the thickness of the thin film due to heterogeneous sublimation, and (3) therefore, impossible to obtain a thin film with excellent physical properties. In order to solve these problems, specially designed canisters for solid-state precursors are required to maintain a constant vapor pressure during the deposition process. For these reasons, Mo precursors in the form of a liquid phase are currently being investigated by many researchers. Several organometallic molybdenum complexes such as $Mo(O^tBu)_5$, $Mo(THD)_3$, $Mo(NMe_2)_4$, etc., have been reported so far.[21–24] However, all these Mo precursors are also obtained as solid-state precursors at room temperature.

In this study, we have successfully prepared a new Mo precursor in the liquid phase by a simple ligand exchange reaction using $MoCl_5$ as a starting material. The bidentate ligand, 2,2,6,6-tetramethyl-3,5-heptanedione (THD), in $MoCl_4$(THD)(THF) liquid precursor is selected due to its stable structure. This ligand has been widely used in the preparation of other organo-metallic precursors (e.g., indium, gallium, nickel, etc.), which has been used in various metal/metal composite ALD processes.[25–27] The newly synthesized liquid Mo precursor was employed in an ALD super-cycle process and a pure $MoN_x$ thin film with few impurities (C and O) has been prepared. $MoN_x$ thin films prepared in this study exhibit excellent diffusion barrier characteristics due to good crystallinity and a low impurity concentration.

## II. EXPERIMENT

### A. Synthesis of liquid Mo precursor, MoCl4(THD)(THF)

In an ice bath, 3 g of $MoCl_5$ (11 mmol) and 100 ml of anhydrous THF were added dropwise to flask 1 and dissolved. In an ice bath, 20 ml of anhydrous THF and 2.3 ml of THDH (2,2,6,6-tetramethyl-3,5-heptanedione, 11 mmol) were added to flask 2, and then 4.4 ml of Li-nBu (n-butyllithium solution 2.5M in hexanes 11 mmol) was slowly added and reacted for 30 min. Furthermore, the solution from flask 2 was slowly added to flask 1 using a cannula, and then, the solution reacted under reflux for 18 h. Subsequently, the reaction solution was dissolved using an anhydrous acetonitrile solvent, filtered to remove LiCl, and the filtered solution was heated to 150 °C and placed under vacuum to remove the organic solvent and THDH. After further purification with anhydrous diethyl ether, anhydrous hexane, and anhydrous dichloromethane, any remaining organic solvents were removed using vacuum. Approximately 4.03 g (yield; 80.1%) of dark green liquid was obtained and stored in a glovebox due to its vulnerability to moisture and oxygen.

ICP-AES (Found: Mo, 18.8; C, 35.3%, Calculated: Mo, 19.4; C, 36.4%).

### B. Film deposition

Figure 1 shows a schematic of the chamber in which $MoN_x$ films are deposited. As shown in the figure, $MoN_x$ thin films were deposited using a shower head and a downstream type ALD equipment. A 100 nm $SiO_2$ thin film was used to deposit the $MoN_x$ thin film. To remove organic elements on the $SiO_2$ thin film before deposition, a cleaning process was performed using acetone, methanol, and DI water for 15 min each. In the case of the Si substrate, the substrate used for deposition was a p-type Si wafer in the (100) direction. Before the $MoN_x$ thin film deposition, to remove the native oxide on the substrate, it was cleaned with a hydrofluoric acid solution (1:100, HF:DI water). $MoN_x$ thin film was deposited using a newly synthesized Mo precursor and $NH_3$ and $H_2$ gases. During deposition, the Mo precursor was maintained at 104 °C in an external container and delivered to the chamber using Ar carrier gas at 100 SCCM. The line temperature was maintained at 114 °C to prevent condensation of the precursor. The Ar purge gas flow rate was maintained at 600 SCCM. Ar gas with 99.999% purity was used. $MoN_x$ thin films were deposited at 350–500 °C, and the process pressure was kept constant at 0.5 Torr by using a throttle valve.

The ALD process was carried out in two ways, as shown in Fig. 2. The first method is a general ALD process that proceeds in the order of precursor injection-purge-$NH_3$ reactant

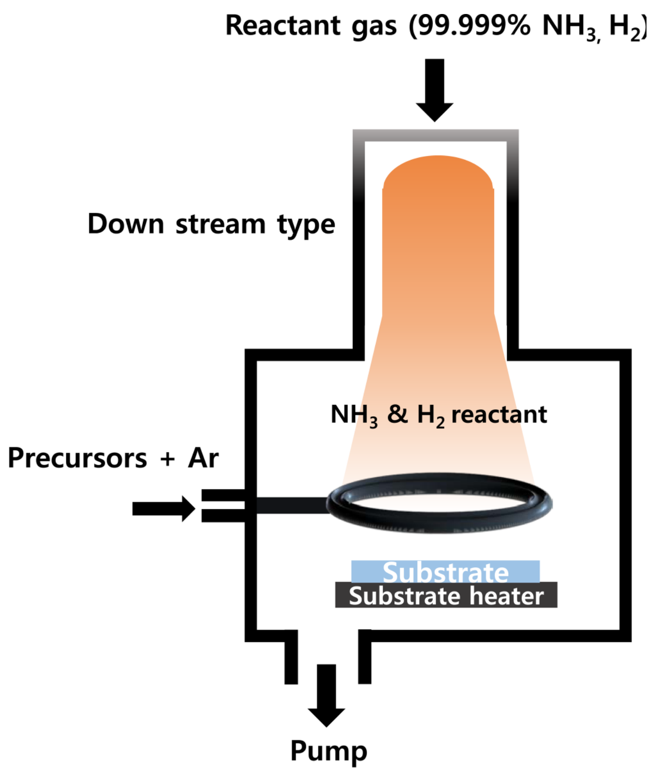


FIG. 1. Simplified schematic of the ALD reactor used for the deposition of $MoN_x$ thin films.

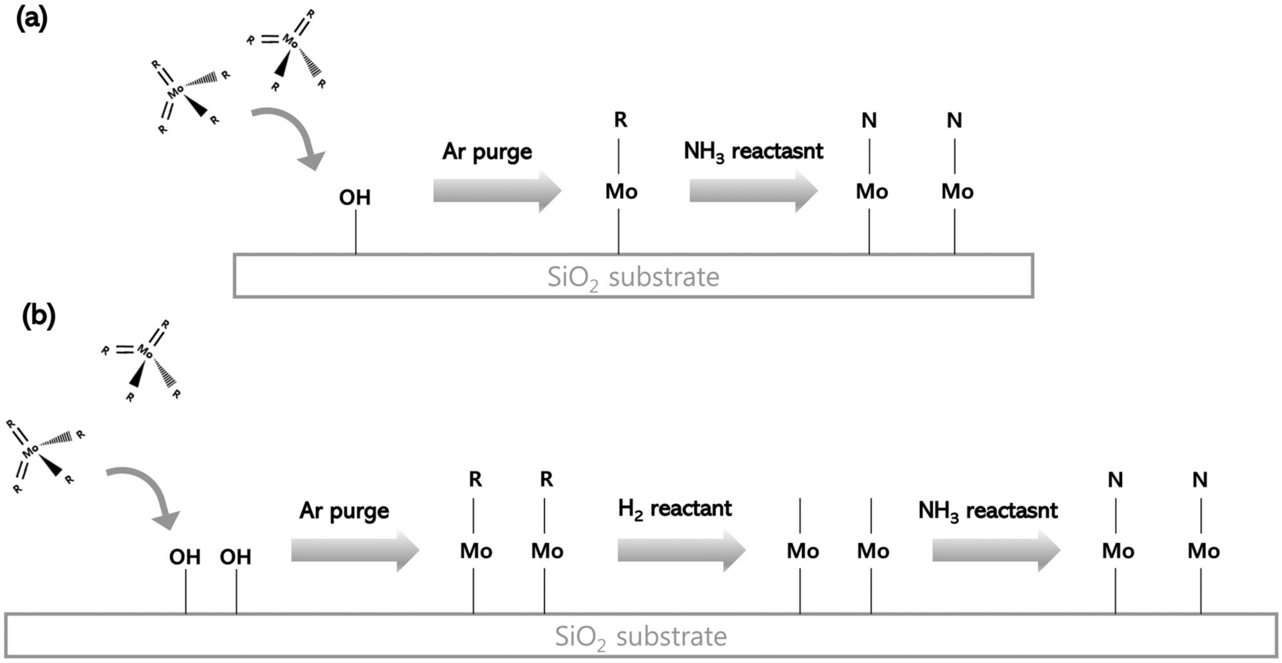


FIG. 2. Simplified schematics of (a) a conventional ALD process and (b) the ALD super-cycle process.

injection-purge [Fig. 2(a)]. The second deposition method is an ALD super-cycle process that proceeds in the order of precursor injection-purge-$H_2$ reactant injection-purge-NH3 reactant injection-purge [Fig. 2(b)]. The reason for using the ALD super-cycle is to effectively reduce the precursor ligand using $H_2$ gas and then deposit the $MoN_x$ film using $NH_3$ gas. The $NH_3$ gas and $H_2$ gas used were 99.999% pure. The precursor injection time was 10 s to supply a sufficient amount of precursor, and the $NH_3$ and $H_2$ gases were supplied for 60 and 20 s, respectively, for sufficient reaction. The process was carried out at 350–500 °C. Since the BEOL



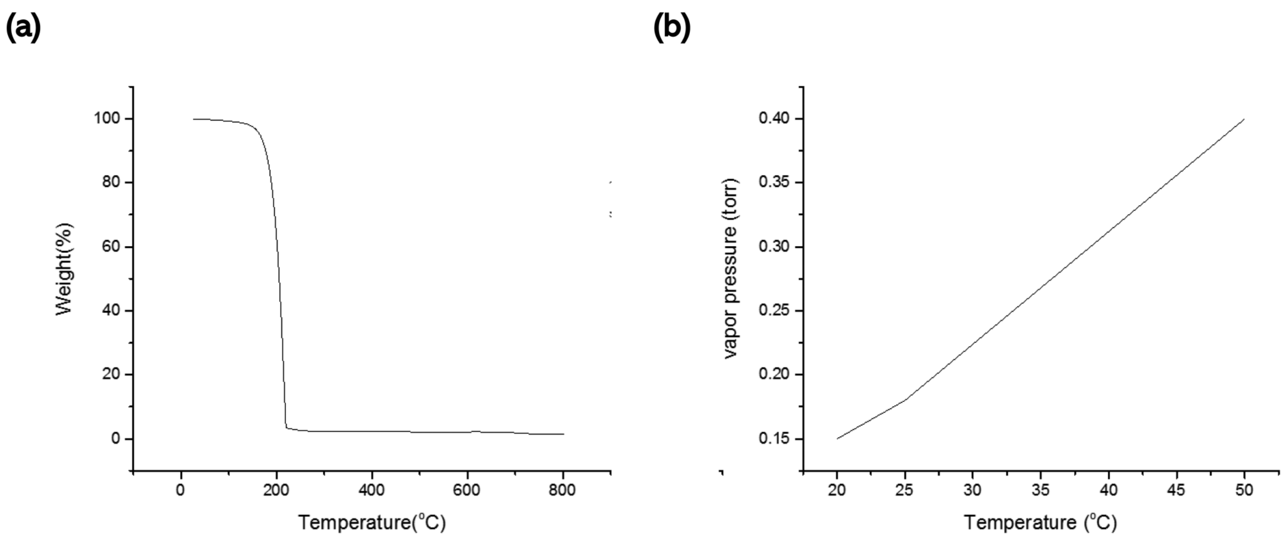


FIG. 3. (a) TGA curve and (b) vapor pressure curve of the synthesized $MoCl_4$(thd)(THF).

**TABLE I.** Boiling points and molecular formula for various Mo precursors.

| Precursor | Chemical formula | Boiling point (°C) | Reference |
|---|---|---|---|
| $Mo(acac)_3$ | $Mo(CH_3C(C{=}O)CH(C{=}O)CH_3)_3$ | 280 (decomposed) | 28 |
| $Mo(thd)_3$ | $Mo(CH_3)_3C(C{=}O)CH(C{=}O)C(CH_3)_3)_3$ | 225 | 22 |
| $MoO_2(thd)_2$ | $Mo({=}O)_2(CH_3)_3C(C{=}O)CH(C{=}O)C(CH_3)_3)_2$ | 275 | 29 |
| $MoCl_4(thd)(THF)$ | $MoCl_4(CH_3)_3C(C{=}O)CH(C{=}CH_3)_3)(O(CH_2CH_2)_2)$ | 200 | 9 |

(back end of the line) process allows up to a process temperature of 500 °C, the thin film deposition temperature was set up to 500 °C in this experiment.

To analyze the thickness of $MoN_x$ thin films, x-ray reflectivity (XRR, Rigaku) was used. XPS (x-ray photoelectron spectroscopy, K-Alpha +) was used to ascertain the bonding between elements inside the thin film. GIXRD (grazing incidence x-ray diffraction, Rigaku) was used to determine the crystallinity of the thin films, and AES (Auger electron spectroscopy, PHI-800Xi) was used to evaluate the atomic concentrations in $MoN_x$ thin films. Sheet resistance was analyzed through a 4-point probe (Sr 5000) measurement, and resistivity was calculated according to the thickness and sheet resistance of the thin film. UPS was used to determine the work function of the $MoN_x$ thin films.

## III. RESULT AND DISCUSSION

The synthesis of the liquid Mo precursor was carried by the ligand displacement of $MoCl_5$ with lithium-2,2,6,6-tetramethyl-3,5-heptanedionate (Li-THD), as a bidentate ligand. The ligand displacement reaction proceeded readily at a reflux condition with the generation of insoluble reaction by-products, (LiCl), in organic solvents. The newly synthesized Mo precursor is easily miscible (soluble) in most of the organic solvents, such as acetonitrile, diethyl ether, and hexane.

The thermal properties of $MoCl_4$(THD)(THF) were evaluated by comparing with $Mo(THD)_3$, $Mo(acac)_3$ (acac=acetylacetonato), and $MoO_2(THD)_2$, which have the same or similar ligands as that in the synthesized precursor. First, it was found that $Mo(acac)_3$ decomposes without evaporation. It is initially decomposed by 50% at 100–275 °C, and then, the final decomposition of 13.7% occurred at 275–550 °C, yielding a residue of 36.3%.[28] In the cases of $MoO_2(THD)_2$ and $Mo(THD)_3$, they both evaporate without any residue. For $Mo(THD)_3$, it evaporates 95% at 150–265 °C and then 3% at 265–300 °C.[22] Meanwhile, $MoO_2(THD)_2$ evaporates 99% at 150–300 °C.[29,30] On the other hand, the newly synthesized Mo precursor, $MoCl_4$(THD)(THF), exhibits the evaporation of 99% at 150–220 °C, which corresponds to the markedly reduced boiling point compared to those of other Mo precursors described above. As can be seen in Fig. 3(a), the evaporation window is also quite narrow, which represents a great advantage of this Mo precursor for ALD and CVD processes. As shown in Fig. 3(b), the vapor pressure curve for $MoCl_4$(THD)(THF) exhibits a vapor pressure of 0.4 Torr at 50 °C. Table I presents the boiling points and molecular formula for various Mo precursors discussed earlier.

Next, the $MoN_x$ thin films were deposited using the synthesized Mo precursor. Figure 4 shows AES data of $MoN_x$ films deposited at 350 and 500 °C. Our research team injected the precursor for 10 s when depositing the $MoN_x$ film, because the GPC was saturated from the precursor feeding time of 8 s (~1.1 Å), and the precursor was injected for 10 s to sufficiently inject the Mo precursor. As shown in the AES data [Fig. 4(a)] of the $MoN_x$ film deposited at 350 °C, the oxygen and carbon concentration in the thin film is high. Since the deposition was carried out at a low temperature, the precursor and reactant did not react sufficiently, and the ligand was not sufficiently reduced, leaving impurities in the thin film. However, as shown in the AES data [Fig. 4(b)] of the $MoN_x$ film deposited at 500 °C, the oxygen and carbon contents in the thin film are reduced by about 10 and 5%, respectively, compared to the $MoN_x$ film deposited at 350 °C. This reduction is due to the high thermal energy supplied during deposition at 500 °C; the reaction



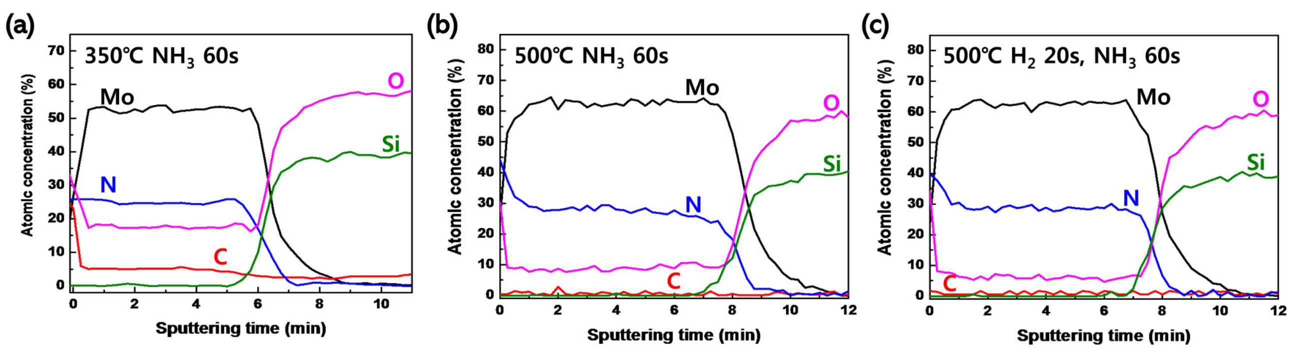


**FIG. 4.** (a) AES data of $MoN_x$ films deposited using the conventional ALD process at 350 °C, (b) AES data of $MoN_x$ films deposited using the conventional ALD process at 500 °C and (c) AES data of $MoN_x$ films deposited using the ALD super-cycle process at 500 °C.

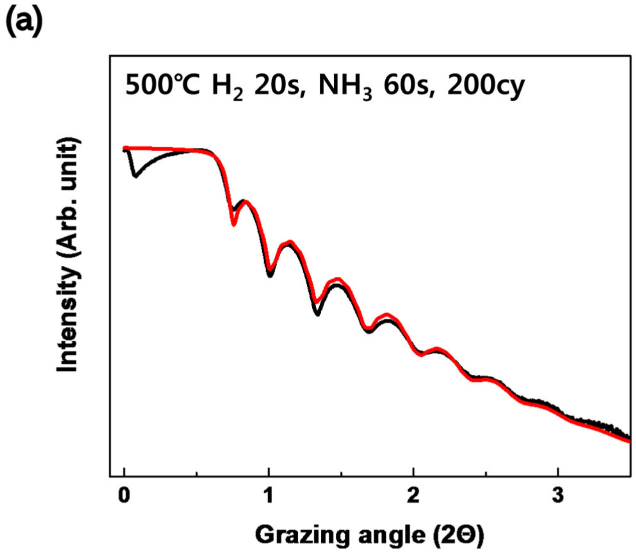


**FIG. 5.** (a) Raw XRR data of the $MoN_x$ film with a curve fit and (b) the thickness of the $MoN_x$ thin film as a function of the number of ALD cycles.

between the precursor and reactant becomes active, and the ligand is efficiently removed. However, even when the $Mo_2N$ film is deposited at 500 °C, oxygen still constitutes about 10% of the thin film. Oxygen in the metal film increases the resistivity of the thin film. Therefore, $MoN_x$ film was deposited using $H_2$ gas and an ALD super-cycle to lower the oxygen content of the thin film. As shown in Fig. 2, when using $H_2$ gas, the ligand is removed by the $H_2$ gas before reacting with $NH_3$ gas. Therefore, fewer impurities remain in the thin film. As shown in the AES data [Fig. 4(c)] of the $MoN_x$ film deposited using the ALD super-cycle process, the oxygen concentration in the thin film is about 5–6%, which is reduced by about 4–5% compared to the $MoN_x$ film deposited using the general ALD process. Through the AES data, it was confirmed that a $MoN_x$ film can be deposited using the ALD super-cycle at a substrate temperature of 500 °C. Therefore, the substrate temperature was maintained at 500 °C and the deposition of $MoN_x$



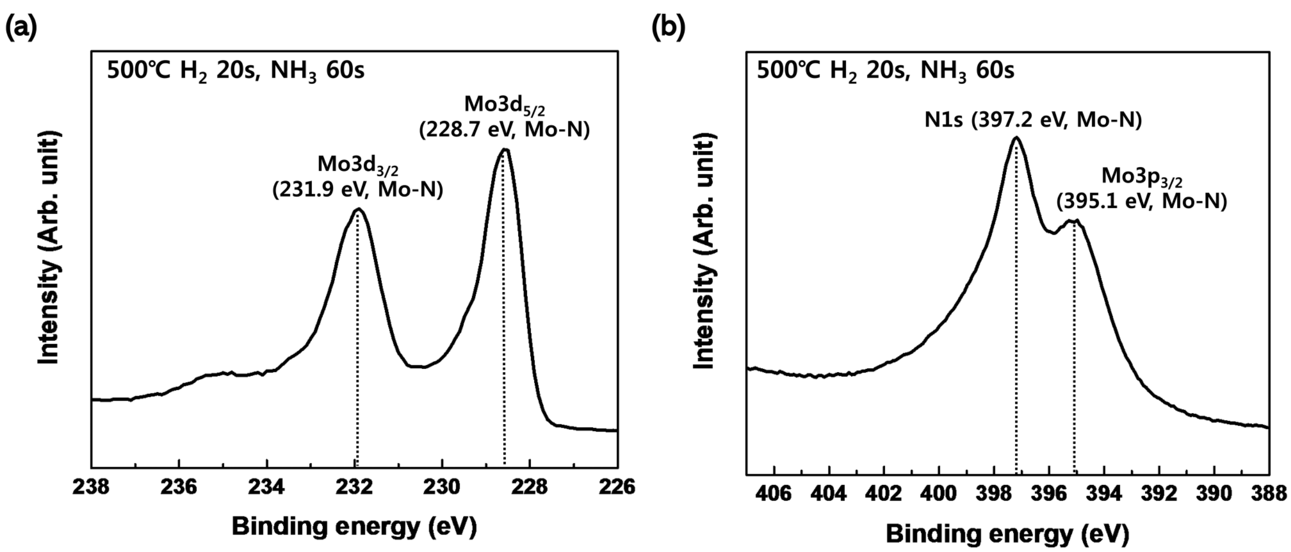


**FIG. 6.** XPS data of (a) Mo3d and (b) N1s and Mo3p of an $MoN_x$ thin film deposited by ALD super-cycles.

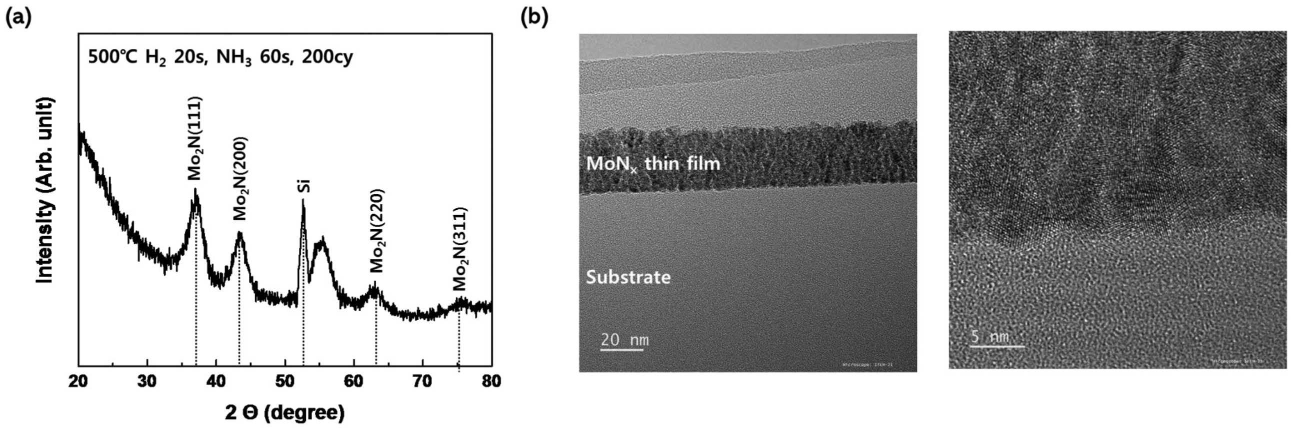


**FIG. 7.** (a) GIXRD data and (b) HR-TEM image of $MoN_x$ thin films deposited by ALD super-cycle.

films was achieved using the ALD super-cycle process for subsequent experiments.

The thickness of the $MoN_x$ film deposited using the ALD super-cycle process at 500 °C was measured after several cycles of deposition to check if a self-limited reaction occurred. Figure 5(a) shows raw XRR data of a $MoN_x$ film deposited using an ALD super-cycle and a substrate temperature of 500 °C along with a curve fit. As the figure shows, the raw data and the curve fit are in good agreement. Therefore, we conclude that the thicknesses obtained through XRR analysis are reliable. Figure 5(b) shows the calculated thicknesses after deposition of 50–300 of the ALD super-cycles. It can be seen that the thin film thickness increases linearly with the number of ALD cycles. Therefore, it is confirmed that the $MoN_x$ thin film was deposited by an ALD reaction. There are cases in which the pulsed CVD process has also yielded the same results. However, according to the previously published papers (S. C. Sun, Cat. No. 98EX105. IEEE, 1998), in the case of $MoN_x$ deposited by CVD, crystallinity is secured at 550 °C or higher and it can be confirmed that it has a low resistivity value. Therefore, the $MoN_x$ film deposited in this study has a low impurity content and a low resistivity value; therefore, it is thought to be deposited by ALD.

XPS analysis was performed to confirm the bonding states of Mo and N in the $MoN_x$ thin films. XPS spectra showed that the Mo3d and N1s peaks were the main peaks. As shown in Fig. 6(a), the $Mo3d_{3/2}$ peak binding energy is 231.9 eV and the $Mo3d_{5/2}$peak binding energy is 228.7 eV. Additionally, as shown in Fig. 6(b), the N1s peak binding energy is 397.2 eV and the $Mo3p_{3/2}$ peak binding energy is 395.1 eV. Though the data are not shown, the $Mo3p_{1/2}$ peak was also detected. The binding energies of Mo3d, Mo3p, and N1s in the $MoN_x$ thin film deposited using the newly synthesized precursor and the ALD super-cycle had the same binding energies as in previously reported $MoN_x$ thin films.[11,12]

Figure 7(a) shows GIXRD of a deposited $MoN_x$ thin film. As shown in Fig. 7(a), $Mo_2N$ (111), $Mo_2N$ (200), $Mo_2N$ (220), and $Mo_2N$ (311) peaks are detected from the $MoN_x$ film deposited using the 500 °C ALD super-cycle, indicating excellent crystallinity of the $MoN_x$ film deposited using the newly synthesized precursor and the ALD super-cycle process.[12,13] Figure 7(b) shows an HR-TEM image of the deposited $MoN_x$ thin film. As shown in the TEM image, it can be seen that the $MoN_x$ thin film is deposited on the substrate. Additionally, as shown in the high-resolution TEM image, it can be confirmed that the deposited $MoN_x$ is grown as polycrystalline. It was confirmed that the $MoN_x$ thin film deposited through GIXRD and HR-TEM had excellent crystallinity.

As shown in Fig. 8, we performed a UPS analysis to determine the work function of the $MoN_x$ thin film. The work function was calculated using the following equation:

$$\phi = h\nu - |E_{cut\text{-}off} - E_F|,$$

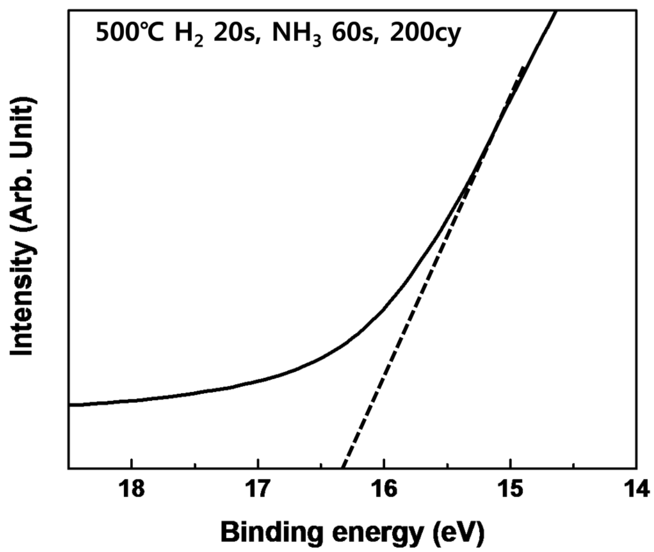


**FIG. 8.** UPS data of $MoN_x$ thin films deposited by ALD super-cycle.

**TABLE II.** Comparison of the resistivities of $MoN_x$ thin films produced from different deposition methods, precursors, and reactants.

| Deposition method | Precursor and reactant | Deposition temp. (°C) | Resistivity ($\mu\Omega$ cm) | Reference |
|---|---|---|---|---|
| PLD | Mo target and $N_2$ gas | 25–500 | 145–1500 | 13 |
| CVD | $Mo(NCH_3)_2)_4$ and $NH_3$ | 200–400 | 10000 | |
| ALD | $Mo(CO)_6$ and $NH_3$ plasma | 180–230 | 3000–4500 | 12 |
| | New synthesized Mo precursor and $H_2$, $NH_3$ gas | 500 | 1200 | — |

where $\Phi$ is the work function of the deposited $MoN_x$ thin film, $E_{cut\text{-}off}$ is the inelastic high binding energy cutoff, and $E_F$ is the Fermi edge. The incident photon energy was provided by a He I source (21.2 eV). The difference between the cut-off energy and the Fermi edge was measured to be approximately 16.3–16.4 eV. Therefore, the work function of the $MoN_x$ thin film was calculated to be 4.9–5.0 eV, which is well within the range of 4.5–5.2 eV reported for $MoN_x$ films in prior studies.[31] Additionally, it was confirmed that the deposited $MoN_x$ thin films had work functions suitable for PMOS devices.[32,33]

As shown in Table II, the resistivity of $MoN_x$ films deposited using our newly synthesized Mo precursor and the ALD super-cycle process is comparable to $MoN_x$ films deposited using other deposition methods and precursors. In fact, it is evident that the resistivity of the $MoN_x$ film deposited using the newly synthesized Mo precursor and the ALD super-cycle process is lower than that of the $MoN_x$ film deposited using other deposition methods and existing precursors. The reason for the low resistivity is that our $MoN_x$ thin film contains fewer impurities, such as oxygen and carbon, and exhibits crystallinity, as shown in the GIXRD data. Additionally, as a result of the resistivity analysis according to the thickness of the $MoN_x$ film, the resistivity of the thin film was maintained at 1200–1300 $\mu\Omega$ cm up to 5 nm, but it was confirmed that the resistivity of the thin film increased to more than 4000 $\mu\Omega$ cm when it was 3 nm. In conclusion, depositing a $MoN_x$ film with excellent characteristics using our newly synthesized precursor and the ALD super-cycle process should be useful for next-generation memory and logic devices.

In order to confirm the diffusion barrier properties of the deposited $MoN_x$ film, a postannealing process was performed on a Cu (100 nm)/$MoN_x$ (5 nm)/sub stack. To examine the diffusion barrier properties, a $MoN_x$ film was deposited on a Si wafer and XRD analysis was performed. Figure 9(a) shows a schematic of the annealing process after the Cu (100 nm)/$MoN_x$ (5 nm)/sub stack is fabricated. To evaluate the diffusion barrier properties of the deposited $MoN_x$ film, 5 nm of the $MoN_x$ film was deposited on a substrate, and 100 nm of Cu was deposited on the top of $MoN_x$ using a sputtering process (Ar 100 W plasma, Cu target). Then, the annealing process was performed in a vacuum atmosphere for 5 min. In the current memory device manufacturing process, since the process is carried out at 600 °C or less, the diffusion barrier test was carried out up to 600 °C. As XRD was used to evaluate the diffusion barrier properties of the WCN thin film in previously published papers,[14] this study also conducted the evaluation of diffusion barrier properties using XRD. As shown in Fig. 9(b), XRD analysis shows that only a Cu peak is detected from the Cu (100 nm)/$MoN_x$ (5 nm)/Si wafer stack for which the annealing process was performed at 500 and 600 °C, and the $Cu_3Si$ peak—which appears when Cu diffuses into the Si wafer—was not detected.[34,35]

Additionally, TEM analysis was performed to confirm whether the structure of Cu/$MoN_x$/substrate was well maintained



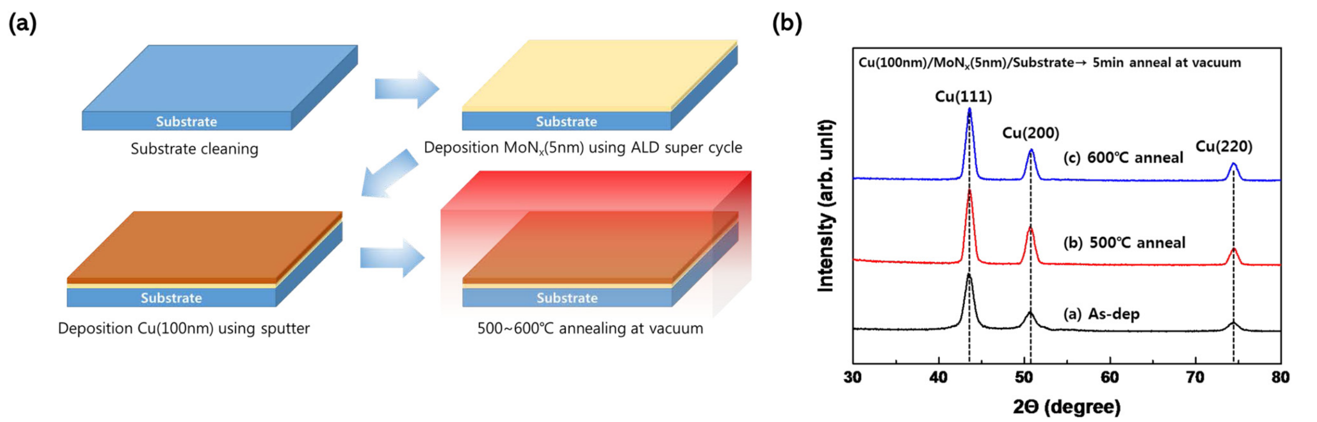


**FIG. 9.** (a) Schematic of deposition Cu/$MoN_x$/sub-structure and annealing process and XRD data of (a) As-dep Cu(100 nm)/$MoN_x$ (5 nm)/sub stack, (b) Cu(100 nm)/$MoN_x$ (5 nm)/sub stack at 500 °C, (c) Cu(100 nm)/$MoN_x$ (5 nm)/sub stack annealed at 600 °C.

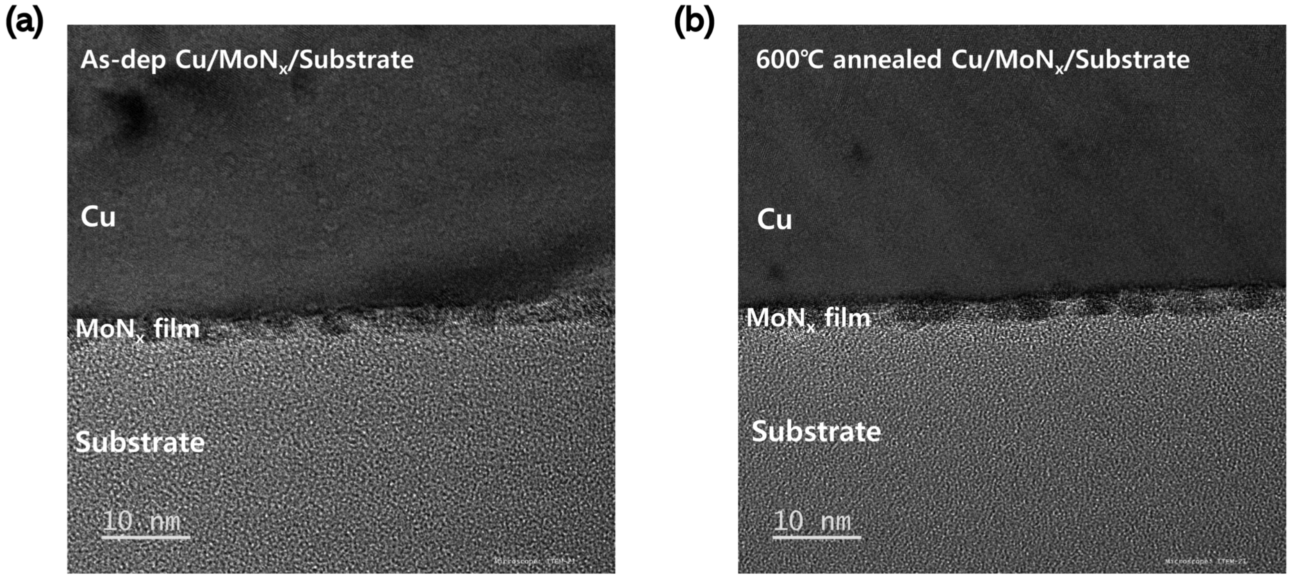


**FIG. 10.** TEM image of (a) as-dep Cu/$MoN_x$/substrate and (b) 600 °C annealed Cu/$MoN_x$/substrate.

before and after the annealing process. In general, a diffusion barrier thin film is deposited on a dielectric film. Therefore, the $MoN_x$ thin film was deposited on the $SiO_2$ wafer to check whether the thin film is maintained after the annealing process. As shown in Fig. 10, It could be confirmed that the Cu/$MoN_x$/sub structure was well maintained before and after the annealing process, and it was confirmed that the $MoN_x$ film was deposited to about 5 nm. The XRD and TEM data confirm that the $MoN_x$ thin film performs excellently as a diffusion barrier up to 600 °C.

## IV. SUMMARY AND CONCLUSIONS

In this study, we have successfully prepared a new Mo precursor by a simple ligand exchange reaction using $MoCl_5$ as a starting material. The bidentate ligand, 2,2,6,6-tetramethyl-3,5-heptanedione (THD), in $MoCl_4$(THD)(THF) liquid precursor is selected due to its stable structure as well as remarkable volatility characteristics. Liquid Mo precursor with a low boiling point of 200 °C has been synthesized and employed in an ALD super-cycle process. Deposition experiments showed that this new precursor is suitable for the ALD process. Deposition using the newly synthesized precursor and ALD processes showed that a superior $MoN_x$ film was deposited when the ALD super-cycle process was used rather than the conventional ALD process. In addition, the $MoN_x$ film deposited using the ALD super-cycle process has low resistivity, due to few impurities and excellent crystallinity, and can serve as a diffusion barrier up to 600 °C. In conclusion, the $MoN_x$ thin film deposited using the newly synthesized precursor and the ALD super-cycle has a low resistivity and can serve as a diffusion barrier even at high temperatures, making it a strong candidate for use in next-generation memory devices.

## ACKNOWLEDGMENTS

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

B.K. and S.L. contributed equally to this work.

**Byunguk Kim:** Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (lead); Visualization (equal); Writing – original draft (lead); Writing – review & editing (equal). **Sangmin Lee:** Investigation (lead); Methodology (lead); Visualization (lead); Writing – original draft (lead). **Taesung Kang:** Investigation (supporting); Methodology (supporting); Validation (supporting); Visualization (supporting). **Sunghoon Kim:** Data curation (supporting); Formal analysis (supporting); Investigation (supporting); Validation (supporting). **Sangman Koo:** Conceptualization (equal); Project administration (equal); Supervision (equal). **Hyeongtag Jeon:** Project administration (equal); Supervision (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available within the article.

## REFERENCES

[1]B. Kim *et al.*, Appl. Surf. Sci. **541**, 148482 (2021).
[2]B. Kim *et al.*, J. Alloys Compd. **857**, 157931 (2021).
[3]W. Zhang, S. H. Brongersma, O. Richard, B. Brijs, R. Palmans, L. Froyen, and K. Maex, Microelectron. Eng. **76**, 146 (2004).
[4]D. Gall, J. Appl. Phys. **119**, 085101 (2016).
[5]M. J. Ha, H. Kim, J. H. Choi, W. H. Kim, T. J. Park, B. Shong, and J. H. Ahn, Chem. Mater. **34**, 2576 (2022).
[6]S. K. Kim, J. H. Han, G. H. Kim, and C. S. Hwang, Chem. Mater. **22**, 2850 (2010).
[7]T. Park, J. Lee, J. Park, H. Jeon, and H. Jeon, Jpn. J. Appl. Phys. **52**, 05FB05 (2013).
[8]S. K. Kim, S. Y. Lee, S. W. Lee, G. W. Hwang, C. S. Hwang, J. W. Lee, and J. Jeong, J. Electrochem. Soc. **154**, D95 (2007).
[9]P. Schmitt, V. Beladiya, N. Felde, P. Paul, F. Otto, T. Fritz, A. Tunnermann, and A. V. Szeghalmi, Coatings **11**, 173 (2021).
[10]A. J. M. Mackus, N. F. W. Thissen, J. J. L. Mulders, P. H. F. Trompenaars, M. A. Verheijen, A. A. Bol, and W. M. M. Kessels, J. Phys. Chem. C **117**, 10788 (2013).
[11]T. H. Kim, D. K. Nandi, R. Ramesh, S. M. Han, B. Shong, and S. H. Kim, Chem. Mater. **31**, 8338 (2019).
[12]Y. H. Joo, D. Nandi, R. Ramesh, Y. Jang, J. S. Bae, T. Cheon, and S. H. Kim, J. Alloys Compd. **858**, 158314 (2021).
[13]Y. Jang *et al.*, J. Alloys Compd. **663**, 651 (2016).
[14]B. Kim, Y. Choi, D. Lee, S. Cheon, Y. Byun, and H. Jeon, Nanotechnology **33**, 115701 (2022).
[15]B. Kim, T. Kang, G. Lee, and H. Jeon, Nanotechnology **33**, 045705 (2022).
[16]B. Kim, T. Kang, S. Song, C. Jung, J. Lee, S. H. Cheon, and H. Jeon, Vacuum **199**, 110957 (2022).
[17]B. Kim, H. Park, S. Lee, D. Lee, C. Jung, and H. Jeon, ECS J. Solid State Sci. Technol. **11**, 053009 (2022).
[18]T. H. Kim, D. Nandi, R. Ramesh, S. M. Han, B. Shong, and S. H. Kim, Chem. Mater. **31**, 8338 (2019).
[19]L. Hiltunen, M. Leskela, M. Makela, L. Niinisto, E. Nykanen, and P. Soininen, Thin Solid Films **166**, 149 (1988).
[20]Y. P. Chang, A. L. Hector, W. Levason, G. Reid, and J. Whittam, Dalton Trans. **47**, 2406 (2018).
[21]J. Hillenbrand, M. van Gastel, E. Bill, F. Neese, and A. Furstner, J. Am. Chem. Soc. **142**, 16392 (2020).
[22]M. Mattinen *et al.*, Adv. Mater. Interfaces **4**, 1700123 (2017).
[23]K. Feenan and G. W. A. Fowles, Inorg. Chem. **4**, 310 (1965).
[24]F. Stoffelbach, D. Saurenz, and R. Poil, Eur. J. Inorg. Chem. **10**, 2699 (2001).
[25]D. Pugh, L. G. Bloor, S. Sathasivam, I. P. Parkin, and C. J. Carmalt, Eur. J. Inorg. Chem. **2011**, 1953 (2011).
[26]M. Ottosson and J. O. Carlsson, Chem. Vap. Depos. **15**, 186 (2009).
[27]D. C. Bradley and M. H. Chisholm, J. Chem. Soc. A: Inorg. Phys. Theor. **1971**, 2741.
[28]A. Y. Ledneva, S. B. Artemkina, D. A. Piryazev, and V. E. Fedorov, J. Struct. Chem. **56**, 1021 (2015).
[29]M. Mattinen *et al.*, Mater. Today Chem. **9**, 17 (2018).
[30]Z. Xue, H. Thridandam, H. K. Kaesz, and R. F. Hicks, Chem. Mater. **4**, 162 (1992).
[31]S. Y. Lin and Y. S. Lai, ECS J. Solid State Sci. Technol. **3**, N161 (2014).
[32]C. H. Fu, K. S. Chang-Liao, H. Y. Lu, C. C. Li, and T. K. Wang, Solid State Electron. **54**, 1160 (2010).
[33]C. H. Fu, P. Y. Chien, K. S. Chang-Liao, T. K. Wang, and W. F. Wu, Solid State Electron. **52**, 1512 (2008).
[34]H. Kim, J. Park, H. Jeon, W. Jang, H. Jeon, and J. Yuh, J. Vac. Sci. Technol. A **33**, 05E111 (2015).
[35]T. Laurila, K. Zeng, J. K. Kivilahti, J. Nolarius, and L. Suni, J. Appl. Phys. **91**, 5391 (2022).